RCS Day 3

Receptacle boxes, conductor paths, terminations, and de-energized checks

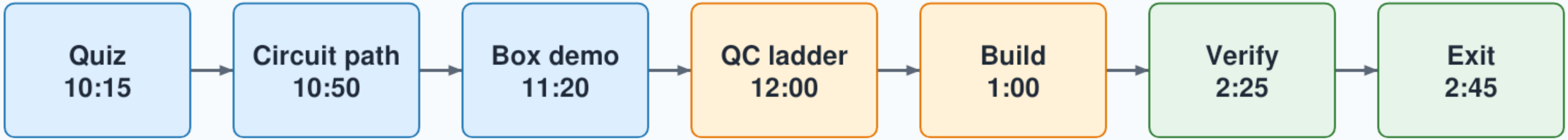
DE-ENERGIZE. TRACE. TERMINATE. VERIFY.

Training mock-ups only. No connection to building power. Every circuit stays isolated.

Today the goal is not to make power. The goal is a safe, inspectable wiring path.

Today's flow

Quiz -> model -> guided practice -> capstone lab -> QC



Morning: trace hot, neutral, and equipment-grounding paths.
Afternoon: mount two boxes and terminate two training receptacles.
End: visual check, tug check, continuity, label, photo, reset.

Today's targets

What successful work must demonstrate

Hot path out

Neutral path back

Grounding fault path

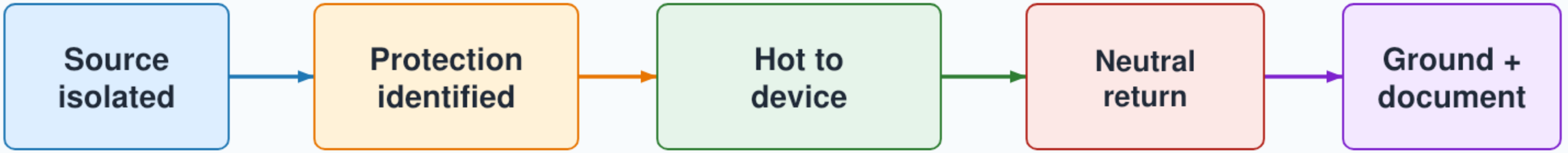
Device boxes

Receptacles + evidence

Trace hot, neutral, and equipment-grounding paths in a training circuit
Lay out two receptacle boxes on the capstone wall
Prepare and terminate conductors on training receptacles
Complete visual, mechanical, and continuity checks while de-energized

The day's path

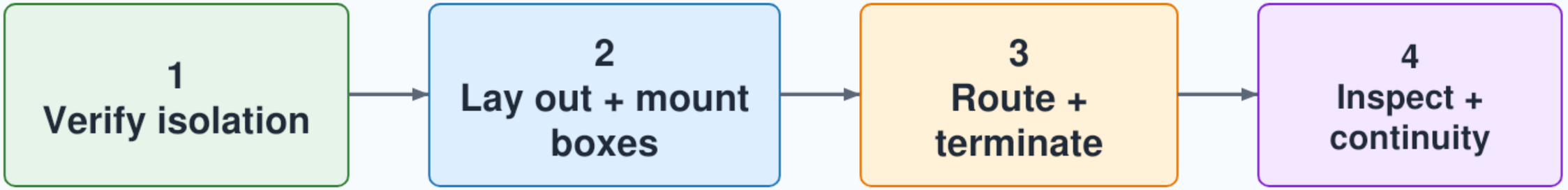
Trace the work before building it



Trace the expected circuit before routing cable. Every student calls the energy state before touching a conductor.

Controlled sequence

Release each stage only after its check



Instructor approves both box locations before drilling. No student test begins until isolation is verified.

Working roles

6 students; every result receives another set of eyes

Builder
routes + terminates

Inspector
visual + tug check

Documenter
logs + photographs

Rotate at the first completed receptacle.
The Inspector checks jacket entry, insulation, copper, and terminal security.
The instructor controls any test source.

Verification checkpoints

Do not trust appearance alone

Isolation proven

Visual inspection

Mechanical tug

Continuity check

Use continuity only on an isolated circuit.
Expected path first; meter setup second; reading third.
A continuity result does not authorize energized operation.

Safety boundaries

The lab begins only inside these controls

No building power

Treat unknown as live

Eyes for drilling

Instructor test control

Keep the mock-up de-energized and separate from premises wiring. Stop for damaged insulation, exposed loose copper, or an unverified energy state.

Professional judgment

Safe work pauses when the evidence conflicts

Good path
isolate -> trace -> terminate -> check

Bad habit
color guess -> connect -> hope

A professional proves the source state and conductor path. Course practice does not authorize unlicensed field electrical work.

Quality criteria

Good physical work leaves visible proof

Mount boxes at the approved layout marks.
Protect cable and bring jacket into the box.
Strip without damaging conductor.
Keep loose copper out of the enclosure.
Visual, tug, continuity, and instructor approval.

Unknown energy state? Stop and verify.

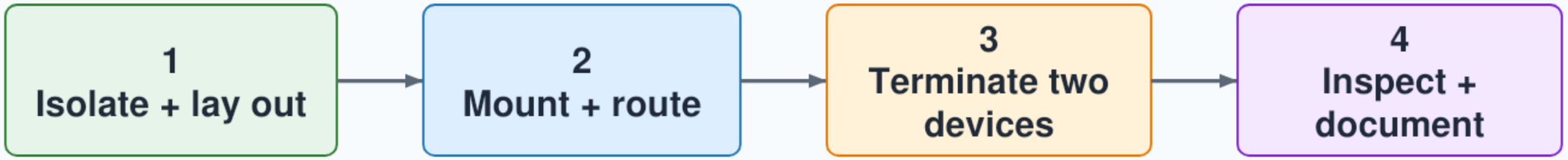
Lab release checklist

The station earns work only after the checks

Training source isolated and identified.
Box measurements recorded.
Drilling layout approved.
Conductors match the training diagram.
Continuity mode used only while isolated.
No connection to premises wiring.

Afternoon lab

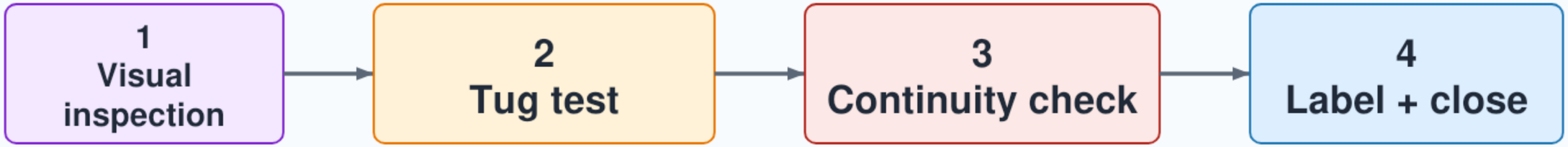
Slow sequence. Clear callouts. No surprises.



Complete two de-energized receptacle terminations on the capstone wall. Instructor approval is the final checkpoint.

Closeout order

End verified, documented, and reset



Record expected and observed results. Leave no exposed conductor, unidentified cable, or energized mystery circuit.

Evidence log

What students write down

Source	isolated + tagged	Initials: _____
Box 1	layout + secure	Measure: _____
Box 2	layout + secure	Measure: _____
Visual	copper + insulation	Pass: _____
Tug	secure terminals	Pass: _____
Continuity	expected path	Result: _____

No verified isolation, no meter use. No complete QC, no lab approval.

Exit ticket

Three checks before dismissal

1. Trace hot, neutral, and equipment-grounding paths.

2. Name the four verification stages used today.

3. State the boundary for continuity testing.

Mock-up isolated. Devices checked. Conductors labeled. Evidence saved.